

Paper 2 - Set 08 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

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1	C	Functions and transformations
2	C	Simultaneous equations and inequalities
3	A	Quadratic functions
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Worked solutions

Question 1 FUN-08 · Functions and transformations · spec MM1.7

For which values of x is $\sqrt{x^2} = -x$?

Answer: C

$\sqrt{x^2} = |x|$, and $|x| = -x$ exactly when $x \leq 0$.

Full method - how and why

The key fact is $\sqrt{x^2} = |x|$, not x . The question is then about when $|x| = -x$.

1 Rewrite the left-hand side: $\sqrt{x^2} = |x|$.

why: The square-root symbol always means the **non-negative** root, so the output cannot be negative even when x is.

2 The equation becomes $|x| = -x$.

why: Both sides must therefore be ≥ 0 , which already forces $-x \geq 0$.

3 $|x| = -x$ exactly when $x \leq 0$ (for $x < 0$, $|x| = -x$; at $x = 0$ both sides are 0).

why: Split into cases $x \geq 0$ and $x < 0$, or recall the definition of $|x|$ directly.

Where it goes wrong

- Answering 'no real x ' on the grounds that a square root cannot equal a negative - but $-x$ is positive when x is negative.
- Excluding $x = 0$ and answering $x < 0$ only.

Takeaway $\sqrt{x^2} = |x|$; and $-x$ is positive when x is negative, so expressions with a minus sign can still be non-negative.

Question 2 SIM-04 · Simultaneous equations and inequalities · spec MM1.4

The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

Answer: C

$x^2 + (x + k)^2 = 8 \Rightarrow 2x^2 + 2kx + k^2 - 8 = 0$. Tangent when $4k^2 - 8(k^2 - 8) = 0 \Rightarrow -4k^2 + 64 = 0 \Rightarrow k = \pm 4$.

Full method - how and why

A line meeting a circle exactly once is a tangency condition: substitute and set the discriminant to zero.

1 Substitute $y = x + k$ into $x^2 + y^2 = 8$: $x^2 + (x + k)^2 = 8$.

why: Eliminate y so the intersections are the roots of a quadratic in x .

2 Expand and collect: $2x^2 + 2kx + (k^2 - 8) = 0$.

why: $(x + k)^2 = x^2 + 2kx + k^2$, so the x^2 terms combine to $2x^2$.

3 Exactly one point means one repeated root: discriminant $= 0$, so $(2k)^2 - 4(2)(k^2 - 8) = 0$.

why: Tangency is precisely the repeated-root case.

4 Simplify: $4k^2 - 8k^2 + 64 = 0 \Rightarrow -4k^2 = -64 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$.

why: Two tangent lines exist - one on each side of the circle - so both signs are genuine.

Where it goes wrong

- Answering $k = \pm 2\sqrt{2}$ by confusing the radius $\sqrt{8}$ with the intercept.
- Giving only $k = 4$; the geometry guarantees a matching tangent with $k = -4$.

Takeaway Tangent to a circle: substitute and set the discriminant to zero - or use 'distance from centre = radius', which gives $\frac{|k|}{\sqrt{2}} = 2\sqrt{2}$ just as fast.

Question 3 QUA-03 · Quadratic functions · spec MM1.3

The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.

Answer: A

$y = a(x - 2)^2 - 3$; at $x = 0$: $4a - 3 = 5$, so $a = 2$. Then $y = 2(x - 2)^2 - 3 = 2x^2 - 8x + 5$, and $a + b + c = 2 - 8 + 5 = -1$ (which is just $y(1)$).

Full method - how and why

Given a vertex, the completed-square form is the natural one. The clever final step is noticing that $a + b + c$ is just $y(1)$.

- 1 Write the completed-square form using the vertex: $y = a(x - 2)^2 - 3$.
why: The vertex (h, k) appears directly in $y = a(x - h)^2 + k$, so two of the three unknowns are already known.
- 2 Use the point $(0, 5)$: $a(0 - 2)^2 - 3 = 5 \Rightarrow 4a = 8 \Rightarrow a = 2$.
why: One extra point is exactly what is needed to pin down the remaining constant.
- 3 Note that $a + b + c$ is the value of $ax^2 + bx + c$ at $x = 1$.
why: Substituting $x = 1$ makes every power of x equal to 1 - a standard shortcut worth remembering.
- 4 Evaluate: $y(1) = 2(1 - 2)^2 - 3 = 2 - 3 = -1$.
why: No need to expand into $2x^2 - 8x + 5$ at all.

Where it goes wrong

- Expanding and then mis-collecting: $2(x - 2)^2 - 3 = 2x^2 - 8x + 5$, and slips in the -8 give 3 or 7.
- Using the vertex as $(-2, -3)$ - in $a(x - h)^2 + k$ the sign of h is already flipped.

Takeaway $f(1)$ is the sum of the coefficients; and a given vertex means you should start from $a(x - h)^2 + k$.

Question 4 LGC-08 · The logic of arguments · spec Arg4

Which of the following is the negation of '**there exists a positive integer n such that $n^2 + n + 41$ is not prime**'?

Answer: C

Negating 'there exists n with not $P(n)$ ' gives 'for all n , $P(n)$ '.

Full method - how and why

Negating an existential statement whose inside is already a negation - so two flips happen.

- 1 Formalise: ' $\exists n$ such that $n^2 + n + 41$ is **not** prime'.
- why:* Note the inner 'not'.
- 2 Negate the quantifier: $\neg \exists n P(n) \equiv \forall n \neg P(n)$.
why: 'There is one' becomes 'there is none', i.e. 'every one fails'.
- 3 Negate the inside: 'not (not prime)' = 'prime'.
why: Double negation cancels.
- 4 Result: **for every positive integer n , $n^2 + n + 41$ is prime.**
why: Exactly one of the original and this is true; in fact this one is false, since $n = 41$ gives $41^2 + 41 + 41 = 41 \times 43$.

Where it goes wrong

- Leaving the inner 'not' in place: 'for every n , $n^2 + n + 41$ is not prime' - that negates the quantifier but not the property.
- Keeping 'there exists' and only flipping the property.

Takeaway $\neg \exists x \neg P(x) \equiv \forall x P(x)$: flip the quantifier **and** the statement inside.

Question 5 ERR-04 · Identifying errors in proofs · spec Err1

Claim: for all real x , if $x^2 > 9$ then $x > 3$.

- I Suppose $x^2 > 9$.
- II Taking square roots of both sides, $x > 3$.
- III This proves the claim.

Which of the following is true?

Answer: A

$x^2 > 9$ gives $|x| > 3$, i.e. $x > 3$ or $x < -3$. The claim itself is false, and line II is where the invalid deduction happens.

Full method - how and why

Taking square roots of an inequality gives a modulus, not a bare inequality. Here the claim itself is false.

- 1 Test the claim: $x = -4$ gives $x^2 = 16 > 9$, but $x = -4 > 3$. The **claim is false**.
why: Settle the truth of the claim first; that tells you an error must exist somewhere.
- 2 Line I is just the supposition - no error.
why: You are allowed to suppose the hypothesis.
- 3 Line II says 'taking square roots, $x > 3$ '. What $x^2 > 9$ actually gives is $|x| > 3$, i.e. $x > 3$ **or** $x < -3$. **First error**.
why: $\sqrt{x^2} = |x|$, not x .
- 4 The offered repair ('say $|x| > 3$ and then deduce $x > 3$ ') is no repair at all, since $|x| > 3$ does **not** give $x > 3$.
why: That option admits the modulus but then makes the same invalid step.

Where it goes wrong

- Accepting the option that says the claim is true - it is not.
- Blaming line III; by then the damage is done.
- Solving $x^2 > 9$ as $-3 < x < 3$, which is the solution of $x^2 < 9$.

Takeaway $x^2 > k^2 \Leftrightarrow |x| > k \Leftrightarrow x > k$ or $x < -k$. Squaring and rooting both lose sign information.

Question 6 PRF-03 · Mathematical proof · spec Prf1

Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

- I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.
- II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.
- III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.
- IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

Answer: A

Assume (II), deduce p even (I), deduce q even (IV), reach the contradiction (III).

Full method - how and why

Ordering a proof by contradiction. The logical skeleton is fixed: assume the opposite, deduce consequences, hit the contradiction.

- 1 The assumption must come first: II sets up $\sqrt{2} = \frac{p}{q}$ in lowest terms.
why: Nothing can be deduced before the supposition is made, and 'no common factor' is the condition that will later be contradicted.
- 2 Next I: squaring gives $p^2 = 2q^2$, so p^2 is even, so p is even, $p = 2k$.
why: It depends on II, and it introduces k , which IV uses.
- 3 Then IV: substituting $p = 2k$ gives $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.
why: IV cannot precede I because it uses k , which I defines.
- 4 Finally III: p and q are both even, contradicting 'no common factor'. Hence $\sqrt{2}$ is irrational.
why: The conclusion must come after **both** parities are established. Order: II, I, IV, III.

Where it goes wrong

- Placing III before IV - the contradiction needs q even too, so it cannot come earlier.
- Starting with I, which refers to p and q before they have been introduced.

Takeaway Order a proof by dependency: each line may only use things already defined or deduced.

Question 7 GRA-05 · Graphs of functions · spec MM8.1

Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

- I It meets the x -axis at exactly one point.
- II It is symmetric about the line $x = 2$.
- III It lies entirely on or above the line $y = 0$.

Answer: H

$|2x - 4| = 0$ only at $x = 2$ (I). $|2x - 4| = 2|x - 2|$ is symmetric about $x = 2$ (II). A modulus is never negative (III).

Full method - how and why

Three properties of a modulus graph. Rewriting $|2x - 4|$ as $2|x - 2|$ makes all three obvious.

- 1 Rewrite: $|2x - 4| = 2|x - 2|$.
why: Factoring the 2 out of the modulus is legitimate because $|2u| = 2|u|$.
- 2 I: the graph meets $y = 0$ when $|x - 2| = 0$, i.e. only at $x = 2$. **True.**
why: A V-shape touches the axis at its single vertex.
- 3 II: $|x - 2|$ depends only on the distance from 2, so the graph is symmetric about $x = 2$. **True.**
why: Replacing x by $4 - x$ leaves the value unchanged.
- 4 III: a modulus is never negative, so the graph lies on or above the x -axis. **True.**
why: $|u| \geq 0$ by definition.
- 5 All three.
why: Match to the option list.

Where it goes wrong

- Thinking the graph crosses the axis twice because $|x^2 - 4|$ -type graphs do - this one is linear inside the modulus.
- Reading the line of symmetry as $x = 4$.

Takeaway $|ax + b| = |a||x + \frac{b}{a}|$: a V-shape with its vertex at the root, never below the axis.

Question 8 X2-30 · Quadratic functions · spec MM1.3

The equation $x^2 + bx + c = 0$ has real coefficients. Which of the following is a **sufficient** condition for it to have two distinct real roots?

Answer: E

$c < 0$ makes the discriminant $b^2 - 4c$ positive. $b^2 \geq 4c$ allows equality (a repeated root); the others do not control the discriminant.

Full method - how and why

'Sufficient' means the condition alone guarantees the conclusion. Translate into the discriminant and check each option **guarantees** positivity.

- 1 Two distinct real roots requires discriminant $b^2 - 4c > 0$ (strictly).
why: Zero discriminant gives a repeated root, not two distinct ones.
- 2 If $c < 0$ then $-4c > 0$, so $b^2 - 4c \geq -4c > 0$. **Guaranteed** - sufficient.
why: $b^2 \geq 0$ whatever b is, so the whole expression is positive.
- 3 $b^2 \geq 4c$ allows equality, which gives a repeated root - not sufficient for **distinct** roots.
why: Read the word 'distinct'; the boundary case ruins it.
- 4 $b \neq 0, c > 0, b > 0$ give no control: e.g. $b = 1, c = 1$ has discriminant $-3 < 0$, no real roots.
why: One failing example shows a condition is not sufficient.

Where it goes wrong

- Choosing $b^2 \geq 4c$ because it looks like the discriminant condition - it is the condition for **at least one** real root.
- Confusing 'sufficient' with 'necessary': $c < 0$ is not necessary ($x^2 - 5x + 6$ has two distinct roots with $c = 6$).

Takeaway Distinct real roots $\Leftrightarrow b^2 - 4c > 0$. A sufficient condition must force that strictly, with no exceptions.

Question 9 LGC-02 · The logic of arguments · spec Arg2

Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?

Answer: C

$x > 2 \Rightarrow x^2 > 4$, so P is sufficient for Q . But $x = -3$ gives $x^2 > 4$ without $x > 2$, so P is not necessary.

Full method - how and why

Necessary/sufficient questions are two separate yes-no checks: does $P \Rightarrow Q$? does $Q \Rightarrow P$?

- 1 Test sufficiency ($P \Rightarrow Q$): if $x > 2$ then $x^2 > 4$. **True**, so P is sufficient.
why: Squaring preserves the inequality for positive numbers.
- 2 Test necessity ($Q \Rightarrow P$): try to break it. $x = -3$ gives $x^2 = 9 > 4$ but $x \not> 2$.
why: One counterexample is enough to destroy an implication; negative numbers are the obvious place to look when squaring.
- 3 So P is sufficient but not necessary.
why: Sufficient = 'enough to guarantee'; necessary = 'can't do without'.

Where it goes wrong

- Getting the direction backwards: ' P is necessary for Q ' means $Q \Rightarrow P$, not $P \Rightarrow Q$.
- Only checking positive x and concluding the two are equivalent.

Takeaway P sufficient for Q means $P \Rightarrow Q$; P necessary for Q means $Q \Rightarrow P$. Check each direction separately, hunting for counterexamples.

Question 10 PRF-16 · Mathematical proof · spec Prf4

A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$** is given with its lines scrambled.

- I Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.
- II Expanding, $a - 2\sqrt{ab} + b \geq 0$.
- III Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

Answer: A

Start from a known truth (III), expand (II), rearrange (I). Starting from the result and working backwards (I, II, III) would need every step to be reversible and is presented in the wrong direction.

Full method - how and why

Direction matters. A proof should start from something known and deduce the result, not start from the result.

- 1 Start with a known truth: $(\sqrt{a} - \sqrt{b})^2 \geq 0$, valid because squares are non-negative and $\sqrt{a}, \sqrt{b}$ exist for $a, b > 0$. That is III.
why: Beginning from an undeniable fact is the safe way round.
- 2 Expand it: $a - 2\sqrt{ab} + b \geq 0$. That is II.
why: $(\sqrt{a})^2 = a$, $(\sqrt{b})^2 = b$, and the cross term is $-2\sqrt{a}\sqrt{b} = -2\sqrt{ab}$.
- 3 Rearrange: $a + b \geq 2\sqrt{ab}$, then divide by 2. That is I.
why: Dividing by the positive number 2 preserves the inequality.
- 4 Order: III, II, I.
why: Known truth $\rightarrow$ algebra $\rightarrow$ required result.

Where it goes wrong

- Choosing I, II, III - that starts by assuming the result. Working backwards is a legitimate way to **find** a proof, but to **present** one every step must be reversed.
- Starting with II, whose inequality has not yet been justified.

Takeaway Present proofs forwards, from an established fact to the conclusion. This one proves AM $\geq$ GM.

Question 11 LGC-23 · The logic of arguments · spec Arg1

Consider the statement: **if a and b are irrational, then $a + b$ is irrational**. Which of the following pairs is a counterexample?

Answer: B

A counterexample needs the hypothesis true (both irrational) and the conclusion false ($a + b$ rational):

$$\sqrt{2} + (-\sqrt{2}) = 0.$$

Full method - how and why

A counterexample must make the hypothesis **true** and the conclusion **false**. Check both halves for each candidate.

1 Hypothesis: both a and b irrational. Conclusion to break: $a + b$ irrational - so we need $a + b$ **rational**.

why: Only a case satisfying the hypothesis can refute the statement.

2 $a = \sqrt{2}$, $b = -\sqrt{2}$: both irrational ✓, and $a + b = 0$, which is rational ✓. **Counterexample.**

why: $-\sqrt{2}$ is irrational: if it were rational, so would its negative be.

3 Reject $a = 1$, $b = -1$: these are rational, so the hypothesis fails and nothing is refuted.

why: A case that does not satisfy the hypothesis says nothing about the implication.

4 Reject $\sqrt{2} + \sqrt{3}$, $\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$ and $\pi + \sqrt{2}$: each sum is irrational, so the conclusion holds and the statement is not broken.

why: These are examples **of** the statement, not against it.

Where it goes wrong

- Picking rational numbers whose sum is rational - the hypothesis is not satisfied.
- Picking irrationals whose sum is irrational - the conclusion is not broken.

Takeaway A counterexample satisfies the 'if' and violates the 'then'. Check both halves every time.

Question 12 ERR-02 · Identifying errors in proofs · spec Err2

Here is a purported proof that $1 = 2$.

- I Let $a = b$, with $a \neq 0$.
- II Then $a^2 = ab$.
- III So $a^2 - b^2 = ab - b^2$, i.e. $(a - b)(a + b) = b(a - b)$.
- IV Dividing both sides by $(a - b)$: $a + b = b$.
- V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

Answer: A

Since $a = b$, $a - b = 0$ and dividing by zero is invalid.

Full method - how and why

The classic ' $1 = 2$ ' fallacy. Somewhere a step is division by an expression that is secretly zero.

- 1 Line I sets $a = b$ with $a \neq 0$ - a legitimate supposition.
why: Nothing wrong yet.
- 2 Line II: multiplying $a = b$ by a gives $a^2 = ab$. Valid.
why: Multiplying both sides by the same quantity is always allowed.
- 3 Line III: subtracting b^2 from both sides and factorising both sides. Valid.
why: $a^2 - b^2 = (a - b)(a + b)$ and $ab - b^2 = b(a - b)$ - correct algebra.
- 4 Line IV divides by $(a - b)$. But $a = b$, so $a - b = 0$: this is division by zero. **First error.**
why: Dividing by zero is undefined, and it is exactly what manufactures the false conclusion.

Where it goes wrong

- Blaming line V, where the absurdity finally appears; the last line is where you **notice** the error, not where it happens.
- Blaming line III - the factorisation is perfectly correct.
- Being reassured by the stated condition $a \neq 0$; it is $a - b$, not a , that vanishes.

Takeaway In every fake proof of this type, look for a division by a factor that the hypotheses force to be zero.

Question 13 PRF-08 · Mathematical proof · spec Prf2

It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?

Answer: D

Contrapositive: if $g(x) \leq 0$ then $f(x) \leq 0$. Since $g(2) = -1 \leq 0$, $f(2) \leq 0$ (it could be 0 or negative).

Full method - how and why

Deduction from an implication. You are given information about the **conclusion**, so the contrapositive is the tool.

- 1 The given rule is $f(x) > 0 \Rightarrow g(x) > 0$.
why: It lets you conclude about g from f , not the other way round - directly, at least.
- 2 Its contrapositive is: $g(x) \leq 0 \Rightarrow f(x) \leq 0$, which is equally valid.
why: Negate and swap: $\neg(g > 0)$ is $g \leq 0$, $\neg(f > 0)$ is $f \leq 0$.
- 3 Apply it at $x = 2$: $g(2) = -1 \leq 0$, so $f(2) \leq 0$.
why: That is all that follows - $f(2)$ could be 0 or negative.

Where it goes wrong

- Concluding $f(2) < 0$ - too strong; the negation of > 0 includes $= 0$.
- Concluding nothing, forgetting that the contrapositive is always available.
- Concluding $f(2) > 0$ by using the implication backwards (affirming the consequent).

Takeaway Information about the conclusion of an implication is used via the contrapositive - and mind the boundary: $\neg(x > 0)$ is $x \leq 0$.

Question 14 CIR-11 · Circles and circle theorems · spec MM3.2

Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point (5, 0) lies on both circles.

Answer: F

Radii 5 and 3, centres 8 apart $= 5 + 3$, so the circles touch externally at one point (I true, II false). That point is (5, 0): $25 = 25$ and $9 + 0 = 9$ (III true).

Full method - how and why

Two circles: compare the distance between centres with the sum and difference of the radii.

- 1 Read off centres and radii: (0, 0) with $r = 5$; (8, 0) with $r = 3$.

why: Both are already in centre-radius form.

- 2 Distance between centres: 8. Compare with $r_1 + r_2 = 8$.

why: Equal to the sum means the circles touch externally - exactly one common point.

- 3 I is therefore true, and II (two intersection points) is false.

why: Two intersections would need $|r_1 - r_2| < d < r_1 + r_2$, i.e. $2 < d < 8$ strictly.

- 4 III: check (5, 0) in both: $25 + 0 = 25$ ✓ and $(5 - 8)^2 + 0 = 9$ ✓. **True.**

why: The touching point lies on the line joining the centres, 5 from one and 3 from the other.

- 5 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming circles that 'meet' must meet twice.
- Forgetting to test III in **both** equations.

Takeaway $d = r_1 + r_2$: touch externally; $d = |r_1 - r_2|$: touch internally; strictly between: two points.

Question 15 NUM-12 · Number, units and estimation · spec M2.3

n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I There is exactly one such value of n .
- II n must be odd.
- III One of n , $n + 2$, $n + 4$ must be divisible by 3.

Answer: H

Among any three numbers of the form n , $n + 2$, $n + 4$ one is a multiple of 3 (they cover all residues mod 3) - III. So one of them must equal 3 itself for all to be prime, giving $n = 3$ only (3, 5, 7) - I. And 3 is odd - II.

Full method - how and why

A prime-triple question. One observation about remainders on division by 3 settles all three statements.

- 1 III: among n , $n + 2$, $n + 4$, the remainders on division by 3 are all different (they differ by 2 and 4, i.e. by 2 and 1 modulo 3), so one of them is a multiple of 3. **True.**
why: Three numbers hitting all three remainder classes must include one divisible by 3.
- 2 I: that multiple of 3 can only be prime if it **equals** 3. Checking the possibilities gives $n = 3$ (3, 5, 7 are all prime); $n = 1$ fails (1 is not prime) and larger cases contain a composite multiple of 3. **True.**
why: So the triple is unique - this is why 3, 5, 7 is famous as the only 'prime triplet' of this form.
- 3 II: $n = 3$ is odd. (Also, any even $n > 2$ would not be prime.) **True.**
why: Both routes give the same verdict.
- 4 All three.
why: Match to the option list.

Where it goes wrong

- Testing a few values, finding only 3, 5, 7, and still doubting uniqueness - the mod-3 argument proves it.
- Thinking n could be 2: then 4 is not prime.

Takeaway Any three numbers spaced 2 apart cover all remainders mod 3, so one is divisible by 3.

Question 16 RSN-17 · Chains of reasoning · spec Prf5

Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?

Answer: A

If Alice is truthful then Ben lies, so Cara is truthful: two truth-tellers, impossible. If Cara is truthful then Alice and Ben both lie; but Alice lying means Ben is truthful: contradiction. So Ben is the truth-teller: Cara lies (indeed Ben is not lying) and Alice lies (Ben is not lying), which is consistent.

Full method - how and why

A truth-teller puzzle. Assume each person in turn is the sole truth-teller and check the whole scenario for consistency.

- 1 Suppose **Alice** is truthful. Then Ben lies, so 'Cara is lying' is false, so Cara is truthful. That makes two truth-tellers - contradiction.
why: Exactly one truth-teller is the constraint; two breaks it.
- 2 Suppose **Cara** is truthful. Then Alice and Ben both lie. But Alice lying means 'Ben is lying' is false, so Ben is truthful - contradiction.
why: Cara's own statement collapses under its consequences.
- 3 Suppose **Ben** is truthful. Then Cara lies, so Ben is not lying ✓ consistent. Alice says 'Ben is lying', which is false, so Alice lies ✓. Cara says both Alice and Ben lie - false, since Ben is truthful ✓.
why: Every statement's truth value matches the assumed roles, with exactly one truth-teller.
- 4 So Ben is the truth-teller.
why: Two cases eliminated, one consistent - the answer is determined.

Where it goes wrong

- Stopping at the first scenario that seems to work without checking every statement.
- Declaring the situation impossible after two failed cases instead of testing the third.
- Forgetting that a liar's statement must come out **false**, which is itself information.

Takeaway Test each candidate exhaustively: a consistent assignment must satisfy every statement **and** the count of truth-tellers.

Question 17 TRG-17 · Trigonometric functions and equations · spec MM4.5

Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

Answer: F

I is the Pythagorean identity. II is false (e.g. $x = \frac{\pi}{2}$: $0 \neq 2$). III: translating $\sin$ by half a period negates it, so III is true.

Full method - how and why

Three identity claims. Two are standard, one is a classic false 'linearity' claim.

1 I: $\sin^2 x + \cos^2 x = 1$ is the Pythagorean identity, true for every x . **True.**

why: It follows from Pythagoras applied to the unit circle.

2 II: test $x = \frac{\pi}{2}$: $\sin \pi = 0$ but $2\sin \frac{\pi}{2} = 2$. **False.**

why: Trig functions are not linear: $\sin(2x) \neq 2\sin x$ (the correct identity, off-specification here, is $2\sin x \cos x$).

3 III: shifting $\sin$ by half a period reflects it: $\sin(x + \pi) = -\sin x$. **True.**

why: Check at $x = 0$: $\sin \pi = 0 = -\sin 0$ ✓; at $x = \frac{\pi}{2}$: $\sin \frac{3\pi}{2} = -1 = -\sin \frac{\pi}{2}$ ✓.

4 I and III only.

why: Match to the options.

Where it goes wrong

- Accepting II because it 'looks like' a factor coming out.
- Doubting III - test it at two values to be sure.

Takeaway Functions are almost never linear: $f(2x) \neq 2f(x)$ in general. Test suspicious identities at $x = \frac{\pi}{2}$.

Question 18 SIM-11 · Simultaneous equations and inequalities · spec MM1.5

Given that $|2x - 3| < 5$, which of the following statements is/are true?

I $-1 < x < 4$

II $x^2 < 16$

III $x > -1$

Answer: H

$-5 < 2x - 3 < 5 \Rightarrow -1 < x < 4$, so I is true and III follows from I. For II: every x in $(-1, 4)$ has $x^2 < 16$ (the largest possible $|x|$ is just under 4), so II is also true.

Full method - how and why

A modulus inequality plus three consequences. Solve the modulus first, then test each statement against that interval.

1 Unpack the modulus: $|2x - 3| < 5$ means $-5 < 2x - 3 < 5$.

why: $|A| < k$ means A lies within k of zero - a double inequality, not two separate cases.

2 Solve the double inequality: add 3 throughout to get $-2 < 2x < 8$, then halve: $-1 < x < 4$. So **I is true**.

why: Whatever you do to one part of a double inequality you do to all three parts.

3 III says $x > -1$, which is half of I. **True**.

why: A weaker statement implied by a true one is automatically true.

4 II: for $-1 < x < 4$, the largest $|x|$ can get is just under 4, so $x^2 < 16$ always. **True**.

why: For squares, what matters is the largest **absolute value** in the interval, not the largest value.

5 All three hold.

why: Match to the option list.

Where it goes wrong

- Treating $|2x - 3| < 5$ as $2x - 3 < 5$ only, losing the lower bound.
- Rejecting II by testing a value like $x = -3$ that is not in the interval at all.

Takeaway $|A| < k \Leftrightarrow -k < A < k$; when squaring an interval, check the end with the larger absolute value.

Question 19 RSN-01 · Chains of reasoning · spec Prf5

A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

Answer: D

Pair up $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$: ten pairs. Choosing 11 numbers forces two from the same pair (pigeonhole); choosing one from each pair gives 10 numbers with no such difference.

Full method - how and why

A pigeonhole question. Build the 'boxes' so that two items in one box give exactly what you want, then answer boxes + 1.

1 Pair the numbers so that each pair differs by 10: $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$.

why: Every number from 1 to 20 lies in exactly one pair, and two numbers differ by 10 precisely when they share a pair.

2 There are 10 pairs, so 10 numbers can be chosen with no two in the same pair - for example 1 to 10.

why: This shows 10 is **not** enough, which is the half of the argument people skip.

3 With 11 numbers and only 10 pairs, two must come from the same pair - and those differ by 10.

why: Pigeonhole: more items than boxes forces a repeat.

4 So the smallest guaranteed n is 11.

why: Boxes + 1.

Where it goes wrong

- Answering 10 - the largest selection that can **avoid** the condition, not the smallest that forces it.
- Answering 2, confusing 'could happen' with 'guaranteed'.
- Forgetting to show 10 is achievable; without it you have not proved 11 is smallest.

Takeaway 'Smallest n that guarantees' = (largest bad selection) + 1. Construct the boxes so a shared box means success.

Question 20 FUN-10 · Functions and transformations · spec MM8.2

$f(x) = x^2 + 1$. Which of the following statements is/are true?

- I $f(f(x)) = x^4 + 2x^2 + 2$ for all x
- II $f(x + 1) = f(x) + 1$ for all x
- III $f(2x) = 4f(x) - 3$ for all x

Answer: F

I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$; true. II: $f(x + 1) = x^2 + 2x + 2 \neq x^2 + 2$; false. III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 1$; true.

Full method - how and why

Three algebraic claims about one function. Each is settled by direct substitution - and the false one needs only a single value of x .

- 1 I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 1 + 1 = x^4 + 2x^2 + 2$. **True.**
why: Substitute the whole of $f(x)$ into f ; expand $(x^2 + 1)^2$ carefully - the cross term $2x^2$ is essential.
- 2 II: $f(x + 1) = (x + 1)^2 + 1 = x^2 + 2x + 2$, whereas $f(x) + 1 = x^2 + 2$. These differ by $2x$. **False.**
why: A single value settles it: $x = 1$ gives $f(2) = 5$ but $f(1) + 1 = 3$.
- 3 III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 4 - 3 = 4x^2 + 1$. Equal for all x . **True.**
why: Compute both sides independently and compare - do not try to 'see' it.
- 4 I and III only.
why: Map the verdicts onto the option list.

Where it goes wrong

- Assuming II is true because functions 'usually' behave additively - almost none do.
- Expanding $(x^2 + 1)^2$ as $x^4 + 1$.

Takeaway Test each claim by substitution; one numerical counterexample is enough to destroy a 'for all x ' statement.